WEEK 6 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | Week Ending: 9th October 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
State model	Completed	Field catalogue for turn state, session state and the draft store. Case History Recall is the persistent-memory use. A reload clears the session, so the case id does not return. The draft stays in drafts.json until the student gives their registration number again. retrieved_context is kept for traces. The next turn searches again.
Memory Design and Data Handling Note	Completed	One use: Case History Recall, reading drafts.json by exact registration number. No new store. The prior-case phrase must be in the current message. The number may be in that message or an earlier student turn. If a case for that number is already on file, the application refuses a second draft. Records stay for the capstone. Deletion is a staff action.
Working memory demonstration	Completed	Merged to main. A prior-case question adds a session note with the case id, queue and issue summary, then the turn continues under the Week 5 limits. Lookup uses only the student's own messages. create_support_case is refused when that registration number already has a draft.
MCP-style interface specification	Completed	Documents the existing create_support_case capability: inputs, outputs, who may call it, and the security boundary. It does not add a tool or a live external system. A call is refused for a blocked topic, a fee decision in the message or only in issue_summary, a turn that has already used three tool calls, and a create_support_case that has already been corrected once in that turn.

Key Engineering Decisions

- Kept the Week 5 session fields and added one memory use, Case History Recall, instead of a third tool or a second file.
- Recall adds a note to the prompt and the turn continues. The model may still request retrieve_documents or create_support_case. The application still applies the Week 5 caps: three model rounds, three tool calls, and create_support_case at most twice in one turn.
- A second draft is refused in code when drafts.json already contains that registration number. The model is not asked to make that decision.
- The registration number is taken from the student's own messages. An example number in an assistant reply is not treated as the student's number.
- The interface specification records create_support_case as it already behaves. Agreement remains the sidebar control. Admissions, grading, discipline and fee decisions stay with staff.

Failures, Challenges and Current Response

- The first demonstration wrote a separate memory file, stopped the turn when it found a prior case, and could treat the sample number in a reply as the student's number. A missing case was described as possibly removed after 90 days. Response: the code on main reads drafts.json, continues the turn, scans only student messages, and does not delete drafts by age.
- The state model first described the memory use as a recommendation, and the interface specification omitted the three-tool-call refusal and a fee decision that appears only in issue_summary. Response: the state model and the memory note on main now match the demonstration.
- issue_summary still accepts a single character. Response: left unchanged, as in Week 5, so the contract and the memory path stay consistent.

Links and Evidence

GitHub repository: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

State model: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/Architecture/Group_M_eve_State_Model.pdf

Memory note: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/Memory_Design_and_Data_Handling_Note.pdf

Memory demonstration: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/src/tools/orchestrate.py

Individual Contribution Summary

Member	Task Owned	Evidence
Nakaye Hildah	Memory Design and Data Handling Note	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/Memory_Design_and_Data_Handling_Note.pdf
Namata Sumayya	State model	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/Architecture/Group_M_eve_State_Model.pdf
Aryampa Joshua	MCP-style interface specification; Week 6 progress report	Specification of create_support_case. The behaviour it records is in src/tools/case_store.py and src/tools/orchestrate.py on main.
Mushabe Moses	Working memory demonstration	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/src/tools/orchestrate.py

Plan for Next Week (Week 7 — Observability)

- Record each student turn so a later reader can see the tools requested, the tools that ran, the stop condition, and whether a draft was saved.
- Keep generated logs out of Git. Leave the three Week 5 traces as the checked-in examples.

- Include one trace where Case History Recall runs and the turn then continues into search or a refused second draft.
- Do not put real registration numbers in a trace. Use the synthetic numbers already used in Week 5.